

Technique	Where to Use	Keywords / Clues	Common Problems	Important Trick
HashMap / HashSet	Fast lookup, counting, duplicates	duplicate, frequency, pair, lookup, count	Two Sum, Group Anagrams, Contains Duplicate	Store visited elements for O(1) lookup
Two Pointers	Sorted arrays, opposite traversal	sorted, pair sum, palindrome, reverse	Two Sum II, 3Sum, Valid Palindrome	Move pointers based on condition
Sliding Window	Contiguous subarray/substring problems	substring, subarray, longest, smallest, contiguous	Longest Substring Without Repeating, Minimum Window Substring	Expand window, shrink when invalid
Binary Search	Efficient searching in sorted data	sorted, search, minimum possible, maximum possible	Search Insert Position, Koko Eating Bananas	Reduce search space by half
Stack	Nested structures, nearest greater/smaller	parentheses, next greater, expression	Valid Parentheses, Daily Temperatures	Use LIFO order
Monotonic Stack	Next greater/smaller problems	nearest greater, histogram	Next Greater Element, Largest Rectangle Histogram	Maintain increasing/decreasing stack
Heap / Priority Queue	Top K, dynamic min/max	top k, kth largest, priority	Top K Frequent Elements, Kth Largest Element	Min heap for K largest

Greedy	Local optimal choice problems	minimum operations, intervals, scheduling	Jump Game, Merge Intervals	Make best local choice
Dynamic Programming (DP)	Repeated subproblems	ways, minimum cost, maximum profit, longest	Climbing Stairs, House Robber, Coin Change	Store previously computed results
Backtracking	Generate all possibilities	all combinations, permutations, subsets	Subsets, Permutations, N-Queens	Choose → Explore → Undo
DFS	Deep traversal in trees/graphs	path, connected, recursive traversal	Number of Islands, Path Sum	Recursive exploration
BFS	Level-order or shortest path traversal	shortest path, level order	Binary Tree Level Order, Rotting Oranges	Queue-based traversal
Graph Algorithms	Connectivity and network problems	graph, dependency, path, cycle	Course Schedule, Number of Provinces	Represent with adjacency list
Topological Sort	Dependency ordering	prerequisites, dependency	Course Schedule	Use indegree + queue
Prefix Sum	Repeated range sum queries	range sum, cumulative sum	Subarray Sum Equals K	$\text{prefix}[R] - \text{prefix}[L-1]$
Cyclic Sort	Numbers in range 1 to n	missing positive, duplicate number	First Missing Positive, Find Duplicate Number	Place x at index x-1
Fast & Slow Pointer	Cycle detection	cycle, linked list loop	Linked List Cycle	Floyd's cycle detection
Union Find (DSU)	Grouping and connectivity	connected components, union	Number of Provinces	Path compression

Kadane's Algorithm	Maximum subarray sum	maximum subarray, contiguous sum	Maximum Subarray	Reset negative running sum
Trie	Prefix searching	dictionary, autocomplete, prefix	Implement Trie, Word Search	Store characters as tree
Segment Tree / Fenwick Tree	Dynamic range queries	update + query ranges	Range Sum Query	Tree-based aggregation
Bit Manipulation	Binary optimization	XOR, single number, bits	Single Number, Counting Bits	XOR cancels duplicates
Matrix Traversal	Grid-based problems	rows, columns, islands	Flood Fill, Word Search	Use direction arrays
Recursion	Divide into smaller subproblems	recursive structure	Tree Traversals, Factorial	Base case + recursive call
Memoization	Optimize recursion	repeated recursion	Fibonacci, DP problems	Cache recursive results
Divide and Conquer	Split problem recursively	merge, partition	Merge Sort, Quick Sort	Divide → Solve → Combine
Sorting	Ordering simplifies logic	sorted order needed	Merge Intervals, Anagrams	Sorting reveals patterns
Frequency Array	Fixed character/number range	lowercase letters, counts	Valid Anagram	Faster than HashMap for fixed range
Deque	Sliding window min/max	window maximum	Sliding Window Maximum	Maintain useful elements only
Binary Search on Answer	Optimization problems	minimum possible, maximize minimum	Koko Eating Bananas	Search answer space

String Matching	Pattern search	pattern matching, substring	KMP, Rabin-Karp	Avoid recomputation
Tree DFS/BFS	Tree traversals	height, depth, diameter	Diameter of Binary Tree	Postorder for bottom-up
Interval Problems	Time ranges and overlaps	intervals, meetings	Merge Intervals, Meeting Rooms	Sort intervals first
Custom Hashing / Ranking	Custom ordering	alien language, custom sort	Alien Dictionary	Map character to rank
Simulation	Follow process exactly	game, robot, movement	Robot Bounded in Circle	Carefully track state
Math / Combinatorics	Counting arrangements	permutations, combinations	Count Unique Digits	Derive formulas carefully
Queue	FIFO processing	scheduling, level processing	Task Scheduler, BFS	Process in arrival order
Linked List Manipulation	Pointer operations	reverse, merge, cycle	Reverse Linked List	Use dummy nodes
In-place Array Manipulation	$O(1)$ space requirement	constant space	Product Except Self, First Missing Positive	Reuse input array